4th Year (Odd Semester)

CSE4111: Parallel Processing and Distributed System
Credits: 3 Contact Hours: 39
Year: Fourth Semester: Odd

Prerequisite: CSE1121: Structural Programming Language, CSE2221: Design and Analysis of Algorithms

Course Type ☒ Theory ☐ Laboratory work ☐ Project work ☐ Viva Voce

Motivation This course is offered for the students to make aware of the potentially vast computing performance improvements that can be obtained with parallelism and exploit this potential when the need arises.

Course Objective:
To make students familiar with multicore programming, shared-memory programming, distributed-memory programming, levels of parallelism, designing efficient parallel algorithm, and programming for massively parallel processor.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **understand** data parallel and task parallel programming pattern to develop a high-performance parallel application.	**Engineering knowledge.** (PO1)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☐ Presentation
CO2	To **design** parallel algorithm for the purpose of exploiting parallel processors	**Design/development of solutions:** (PO3)	Cognitive domain – level 1	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☐ Presentation

Assessment and Marks Distribution:
Students will be assessed on the basis of their overall performance in all the exams, class tests, assignments, and class participation. Final numeric reward will be the compilation of:

Class tests + Assignments due in different times of the semester (20%)
A comprehensive final exam (70%), Total Time: 3 hours.
A class participation mark (10%).

Course Contents:

Introduction: Parallel processing systems and mechanisms, level of parallel processing, parallel vs. non-parallel computing, Multiprogramming and Time sharing, Parallel Computer Structures, parallel processing in uniprocessor system, Parallel processing applications, Speedup and Performance Laws, conditions of parallelism; Bernstein's laws; Parallel Random Access Machines (PRAM); parallel algorithms and complexities.

Parallel Computer Architecture: Advanced processor Technologies, Superscalar and Vector processors, Shared memory architectures, Distributed memory architectures, Design of Linear and Nonlinear Pipeline processors, Multiprocessor System Interconnects; Cache coherence, snoopy-bus and directory-based protocols.

Pipelining and Vector Processing: Principles of Pipelining, Classification of pipelined processors, Instruction and Arithmetic pipeline design, linear and non-linear pipelining, Collision free scheduling in non-linear pipelining, Vector Processing principles, Vector processing requirements, Compound Vector processing, Vectorization and Optimization methods.

Parallel Programming: Parallel Programming models, Parallel Languages and Compilers, Parallel Programming Environments; Distributed Memory programming with MPI; Shared Memory Programming with Pthreads, OpenMP.

Distributed System: Introduction to Distributed System Architectures, Communication in Distributed Systems, Distributed Middleware, Client/Server Design Issues, Inter-process communication, Distributed Objects and Remote Invocation, Virtualization & Code Migration, Naming, Distributed Synchronization & Coordination, Consistency & Replication in Distributed Systems, Fault Tolerance, Distributed Transactions, Security, P2P

Systems; Cloud Computing; Grid Computing.

Text Book:

1. Kai Hwang (Senior Consulting Edition) : **Advanced Computer Architecture Parallelism, Scalability, Programmability,** McGraw Hill.
2. M.J. Quinn : **Parallel Programming in C with MPI and OpenMP,** *McGraw-Hill*

Books Recommended:

1. Peter Pacheco : **An Introduction to Parallel Programming,** *Elsevier Inc.*
2. Tanenbaum and Van Steen. : **Distributed Systems: Principles and Paradigms.** Prentice Hall.

CSE4112: Parallel Processing and Distributed System Lab
Credits: 1 Contact Hours: 26
Year: Fourth Semester: Odd

Prerequisite: CSE1121: Structural Programming Language, CSE2221: Design and Analysis of Algorithms

Course Type ☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce

Motivation This course is offered for the students to achieve implementation details of developing a high performance parallel application through programming and use of parallel hardware.

Course Objective:

To introduce students to give practical experience on programming with multicore CPU, many threads GPU, parallel program using MPI, Pthreads, OpenMP, Java RMI and CUDA C for programming massively parallel processor.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **Implement** parallel program in different levels of parallelism	**Design/development of solutions:** (PO3)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☐ Note book ☐ Presentation
CO2	To **develop** parallel version of sequential program	**Modern tool usage:** (PO5)	Cognitive domain – level 1	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☐ Note book ☐ Presentation

Assessment and Marks Distribution:

Continuous Assessments (CA) (20%)
A comprehensive final exam + Lab note book (70%)
A class participation mark (10%).

Lab Course Contents/ Experiments Lists

1. MPI program that prints greeting from all the processes
2. Parallelizing the Trapezoidal rule for numerical integration using MPI
3. Parallel odd-even transposition sort using MPI
4. Matrix-vector multiplication using Pthreads.
5. Estimate the value of π using Pthreads.
6. A multithreaded linked list
7. Pthreads read-write locks
8. Parallelizing the Trapezoidal rule for numerical integration using OpenMP
9. Parallel odd-even transposition sort using OpenMP

10. Estimate the value of π using OpenMP.
11. Parallelizing the counting sort using OpenMP
12. Parallelizing the reduced solver using OpenMP
13. Remote method for searching a key using Java RMI
14. Kernel function for vector addition using CUDA C

CSE4121: Object Oriented Design and Design Patterns
Credits: 3 Contact Hours: 39
Year: Fourth Semester: Odd

Prerequisite: CSE1221: Object Oriented Programming

Course Type ☒ Theory ☐ Laboratory work ☐ Project work ☐ Viva Voce

Motivation You will discover how to create modular, flexible, and reusable software, by applying object-oriented design principles and guidelines and create interactive applications by incorporating design patterns.

Course Objective: This course takes Java beginners to the next level by covering object-oriented analysis and design. You will discover how to create modular, flexible, and reusable software, by applying object-oriented design principles and guidelines. And, you will be able to communicate these designs in a visual notation known as Unified Modelling Language (UML). And the second part of this course extends object-oriented analysis and design by incorporating design patterns to create interactive applications. Through a survey of established design patterns, you will gain a foundation for more complex software applications. Finally, you will identify problematic software designs by referencing a catalog of code smells.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **analyze** and design the object-oriented model for a problem.	**Problem analysis:** (PO2)	Cognitive domain – level 4	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☐ Presentation
CO2	To **apply** object-oriented modeling principles and design guidelines to create a flexible, reusable, maintainable design.	**Design/development of solutions** (PO3)	Cognitive domain – level 3	☒ Lecture Note ☐ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☐ Final Exam ☐ Assignment ☒ Participation ☐ Presentation
CO3	To **demonstrate** how to use design patterns to address user interface design issues.	**Modern tool usage:** (PO5)	Cognitive domain – level 3	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☐ Final Exam ☒ Assignment ☒ Participation ☐ Presentation
CO4	To **Identify** the most suitable design pattern to address a given application design problem.	**Design/development of solutions** (PO3)	Cognitive domain – level 4	☒ Lecture Note ☐ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☐ Class Test ☐ Final Exam ☒ Assignment ☐ Participation ☒ Presentation

Assessment and Marks Distribution:
Students will be assessed on the basis of their overall performance in all the exams, class tests, assignments, and class participation. Final numeric reward will be the compilation of:

Class tests + Assignments due in different times of the semester (20%)
A comprehensive final exam (70%), Total Time: 3 hours.
A class participation mark (10%).

Course Contents:

Part A: Object-Oriented Design

Object-Oriented Analysis and Design: Good software design begins before coding. After establishing the initial software requirements, design practices involve two main activities: conceptual design and technical design. importance of design and object-oriented thinking, and how to design software using techniques like CRC cards.

Object-Oriented Modeling: Best software design practices have evolved alongside programming languages. Today, all developers should be familiar with abstraction, encapsulation, decomposition, and generalization, which are fundamental principles in object-oriented design. You will learn all of these principles and how they are expressed in Object Oriented Programming and communicated visually in Unified Modelling Language.

Design Principles: Additional design principles will help you to create code that is flexible, reusable, and maintainable. You will learn about coupling and cohesion, separation of concerns, information hiding, and conceptual integrity. You will also learn to avoid common pitfalls with inheritance, and ways to express software behavior in UML.

Part B: Design Patterns

Creational & Structural Patterns: Design patterns help to solve common design issues in object-oriented software. You will learn what they are and how they can be applied. You will learn the creational and structural design patterns. You will continue to learn and practice expressing designs in UML, and code some of these patterns in Java.

Behavioural Design Patterns: You will continue learning useful design patterns and add them to your toolbox. You will learn the behavioral patterns. This will include communicating them in UML and coding them in any OOP Language.

Working with Design Patterns & Anti-patterns: You will learn a design pattern that is very useful for user interfaces: model-view-controller, or MVC. Then you will learn some principles underlying the design patterns, to create software that is flexible, reusable, and maintainable. Finally, you will learn some of the symptoms of bad design, which we call code smells or antipatterns.

Text Book:

1. Grady Booch, Robert A. Maksimchuk, Michael W. Engle, Bobbi J. Young, Jim Conallen, Kelli A. Houston : **Object-Oriented Analysis and Design with Applications**, 3rd Edition, *Addison-Wesley Professional*
2. Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides, Grady Booch : **Design Patterns: Elements of Reusable Object-Oriented Software,** *Addison-Wesley Professional*

CSE4122: Object Oriented Design and Design Patterns Lab
Credits: 1 Contact Hours: 26
Year: Fourth Semester: Odd

Prerequisite: CSE1222: Object Oriented Programming Lab
Course Type ☒ Theory ☐ Laboratory work ☐ Project work ☐ Viva Voce
Motivation You will implement how to create modular, flexible, and reusable software, by applying object-oriented design principles and guidelines and create interactive applications by incorporating design patterns.

Course Objective: This laboratory course takes Java beginners to the next level by covering object-oriented analysis and design. Students will discover how to create modular, flexible, and reusable software, by applying object-oriented design principles and guidelines. And, they will be able to communicate these designs in a visual notation known as Unified Modelling Language (UML). And the second part of this course extends object-oriented analysis and design by incorporating design patterns to create interactive applications. Through a survey of established design patterns, you will gain a foundation for more complex software applications. Finally, students will identify problematic software designs by referencing a catalog of code smells.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **Generate** UML diagram for different components of a software	**Design/development of solutions:** (PO3)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☒ Lab Manual	☐ CA ☐ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO2	To **Breakdown** existing software and rewrite with appropriate design pattern	**Modern tool usage:** (PO5)	Cognitive domain – level 4	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☒ Lab Manual	☐ CA ☐ Final Exam ☒ Assignment ☒ Note book ☒ Presentation

Assessment and Marks Distribution:

Continuous Assessments (CA) (20%)
A comprehensive final exam + Lab note book (70%)
A class participation mark (10%).

Lab Course Contents/List of Experiments:

1. Draw a UML class diagram of an Object-Oriented model for a given topic
2. Create a UML sequence diagram that will show your clients how the system's classes will interact
3. Create a UML state diagram to communicate the state of the system.
4. Draw a UML diagram of a full software project
5. Rewrite a given program based on a given UML diagram using different Structural Design Pattern
6. Write program using different Creational design pattern
7. Write program using different Behavioral design pattern
8. Create a UML class diagram that displays the basic MVC pattern for a web application and write code for that UML diagram.

CSE 4131: Artificial Intelligence
Credits: 3 Contact Hours: 33
Year: Fourth Semester: Even

Prerequisite: CSE2131 Discrete Mathematics, MATH2241 Linear Algebra
Course Type ☒ Theory ☒ Laboratory work ☐ Project work ☒ Viva Voce
Motivation To develop basic knowledge on artificial Intelligence (AI) and its applications.

Course Objective:

This course exposes the most fundamental knowledge on how to realize the intelligent of human behaviors incorporated in a computer system. The student will able to know the fundamental knowledge of AI, different logical expression, knowledge representation, reasoning, resolution, manipulation and a general understanding of AI principles and practice. They understand how to build simple knowledge-based expert systems and various AI search optimization strategies (uninformed, informed algorithms). They also achieve the knowledge on AI programming tools and techniques for real-life problem solving.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **apply** the propositional logic, predicate logic (PL), semantic rules for statements, inference rules for the logical expression	**Engineering knowledge** (PO1)	Cognitive domain – level 3	☒Lecture Note ☒Text Book ☐ Audio/Video ☒Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☒ Assignment ☒ Participation ☐ Presentation
CO2	To **explain** logics of monotonic & non-monotonic, searching techniques, Bayesian probabilistic theorem & algorithms and expert systems to find out the decision for the specific problem.	**Problem analysis**: (PO2)	Cognitive domain – level 5	☒ Lecture Note ☒Text Book ☐ Audio/Video ☒Web Material ☐ Journal paper	☒ Class Test ☒Final Exam ☒Assignment ☒ Participation ☐ Presentation
CO3	To **explain** inductive learning, enforcement learning, neural networks, two-layer & three layer neural net, supervised and unsupervised learning, learning	**Problem analysis**: (PO2)	Cognitive domain – level 5	☒Lecture Note ☒ Text Book ☐ Audio/Video ☒Web Material ☐ Journal paper	☒ Class Test ☒Final Exam ☒Assignment ☒ Participation ☐ Presentation

Assessment and Marks Distribution:
Students will be assessed on the basis of their overall performance in all the exams, class tests, assignments, and class participation. Final numeric reward will be the compilation of:

Class tests + Assignments due in different times of the semester (20%)
A comprehensive final exam (70%), Total Time: 3 hours.
A class participation mark (10%).

Course Contents:
Introduction: History of AI, AI problems, AI Applications, Areas of AI, Intelligence, Knowledge, Types of Knowledge, Agents in AI .

Knowledge Representation: Knowledge representation, Syntax and semantics for Propositional Logic, Predicate Logic (PL), Semantic Rules for statements, Inference Rules, Syntax and Semantics for First Order Predicate Logic (FOPL), Properties of Wffs, Clausal Conversion Procedure, Unification algorithm, Resolution in Propositional Logic, Resolution in Predicate Logic.

Reasoning under uncertainty: Truth Maintenance System (TMS), Logics of Monotonic & Non-monotonic systems, Conceptual dependencies, Bayesian probabilistic theorem for decision.

Searching Techniques& Algorithms in AI: Blind or Uninformed Search, Breadth-First Search, Depth-First search, Informed or Directed search, Best-First Search, Heuristic Search and Hill Climbing search.

Expert systems: Phases in building Expert System, Expert System Architecture, Knowledge Base (KB), Navigational Capability /Inference Engine, Rule Based Expert Systems

Artificial Neural Networks: Biological Neuron, The McCulloch-Pitts model, Neuron Models, Applications of Neural Networks, Two-layer &Three-layer neural net, Supervised learning, unsupervised learning, Deep learning.

Planning and Learning: Planning with state space search, conditional planning, continuous planning, multi-Agent planning, Inductive learning, Reinforcement learning, Explainable AI.

Text Book:

1. D. W. Patterson : **Introduction to Artificial Intelligence and Expert System** *Prentice-Hall of India.*
2. Ivan Nunes da Silva, Danilo Hernane Spatti, Rogerio Andrade Flauzino, Luisa Helena Bartocci Liboni, Silas Franco dos Reis Alves : **Artificial Neural Networks** *Springer international Publishing, Switzerland*

Books Recommended:

1. Elaine Rich, Kevin Knight and Shivashankar B.Nair : **Artificial Intelligence,** *Tata McGraw-Hill, India, 3rd Edition 2009*
2. Patrick Henry Winston : **Artificial intelligence**, *Pearson Education Inc.* 3rd Edition 2011
3. N. P. Padhy : **Artificial Intelligence and Intelligent System**, *Oxford University Press*
4. Bratko, I : **Prolog Programming for Artificial Intelligence,** *Addison Wesley.*
5. Clocksin, W.F. and Mellish, C.S. : **Programming in Prolog: Using the ISO Standard**, *Springer.*

CSE 4132: Artificial Intelligence Lab
Credits: 1 Contact Hours: 26
Year: Fourth Semester: Even

Prerequisite: CSE2131 Discrete Mathematics, MATH2241 Linear Algebra
Course Type ☐ Theory ☒ Laboratory work ☐ Project work ☒ Viva Voce
Motivation To solve the basic problems on artificial Intelligence (AI) and its applications.

Course Objective:
This Lab course exposes on AI programming tools and techniques for real-life problem-solving methods. Students will able to solve problems based on AI and they can make intelligent interface for a system. They can understand how to learn data set and test data with neural network. They also know how to search known and unknown data. Students can perform decision-based learning algorithms.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **apply** the rules to solve the problems, with variables rules, execution rules, input & output predicates and other built in predicates	**The engineer and society** (PO4)	Cognitive domain – level 3	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual	☐ CA ☐ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO2	To **compute** the arithmetical operations compound objects, dynamic and static databases, variables	**Problem analysis**: (PO2)	Cognitive domain – level 5	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual	☐ CA ☐ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO3	To **Demonstrate** expert systems tasks, Backpropagation neural networks & machine learning.	**Modern tool usage**: (PO5)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual	☐ CA ☐ Final Exam ☒ Assignment ☒ Note book ☒ Presentation

Assessment and Marks Distribution:

Continuous Assessments (CA) (20%)
A comprehensive final exam + Lab note book (70%)
A class participation mark (10%).

Lab Course Contents/List of Experiment:

1. Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.
2. Implement a Multilayer Perceptron algorithm without regularization. Update the weights using gradient descent optimizer by implementing the Back propagation algorithm.
3. Classify handwritten digits
4. Classify different images
5. Implement a recurrent neural network to recognize specific words from speech signal.
6. Write a program to implement breath-first search
7. Write a program to implement depth-first search
8. Write a program to implement best-first search
9. Implement Reinforcement learning algorithm
10. Implement software based agent
11. Project on AI problems

Text Book:

1. Carl Townsend : **Introduction to Turbo Prolog,** *Sybex Inc.*
2. Clocksin, W.F. and Mellish, C.S. : **Programming in Prolog: Using the ISO Standard**, *Springer.*
3. Richard L. Halterman : **Fundamentals of Python Programming,** *Southern Adventist University*
4. Paul Deitel : **Intro to Python for Computer Science and Data Science: Learning to Program with AI, Big Data and The Cloud, Global Edition,** *Pearson*

Books Recommended:

1. Brian Heinold : **A Practical Introduction to Python Programming**, *Department of Mathematics and Computer Science, Mount St. Mary's University, ©2012 Brian Heinold*
2. D. W. Patterson : **Introduction to Artificial Intelligence and Expert System** *Prentice-Hall of India.*
3. Patrick Henry Winston : **Artificial intelligence**, *Pearson Education Inc.* 3rd Edition 2011

CSE 4141: Microcontroller Interfacing
Credits: 3 Contact Hours: 39
Year: Fourth Semester: Odd

Prerequisite: CSE2111: Digital System Design, CSE3231: Microprocessor and Microcontroller
Course Type ☒ Theory ☐ Laboratory work ☐ Project work ☐ Viva Voce
Motivation To develop hardware knowledge and programming skills on STM32F4xx microcontroller interfacing

Course Objective:
The main objective of this course is to provide knowledge on basics of interfacing techniques, interfacing devices and finally, to make the student understand on critical programming techniques for STM32F4xx microcontroller so that they can develop engineering skills in designing real world applications.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **identify** the basics knowledge required for interfacing, interfacing devices	**Engineering knowledge** (PO1)	Cognitive domain – level 6	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☒ Presentation
CO2	To **apply** knowledge for real world application using STM32F4xx microcontroller	**Design/Development of Solutions** (PO3)	Cognitive domain – level 4	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☒ Presentation
CO3	To **explain** the use of STM32F4xx microcontroller for real world applications.	**Problem Analysis** (PO2)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☒ Presentation

Assessment and Marks Distribution:
Students will be assessed on the basis of their overall performance in all the exams, class tests, assignments, and class participation. Final numeric reward will be the compilation of:
Class tests + Assignments due in different times of the semester (20%)
A comprehensive final exam (70%), Total Time: 3 hours.
A class participation mark (10%).

Course Contents:

Peripheral I/O and memory mapped I/O, Interfacing with external memory, microprocessor-controlled data transfer and peripheral controlled data transfer, commercial AD and DA converter, ADC 0809, Flash ADC, ADC 0820. 8255 Programmable peripheral interfaces, its different mode of operation. Application of 8259, 8257

STM32F4xx microcontroller architecture, Memory organization, Memory map, CRC calculation unit, Power control (PWR).

STM32 core peripherals and programming, MPU access permission attributes, MPU mismatch, Updating an MPU region, MPU design hints and tips

Nested vectored interrupt controller (NVIC), Level-sensitive and pulse interrupts, NVIC design hints and tips, NVIC register map. System control block (SCB), SysTick timer (STK).

Introduction of different types of Timers, General-purpose timers (TIM2 to TIM5), General-purpose timers (TIM9 to TIM14) of STM32F4xx microcontroller.

Architecture and Programming of Analog-to-digital converter (ADC) of STM32F4xx microcontroller.

Concepts and programming of Universal synchronous asynchronous receiver transmitter (USART), Serial peripheral interface (SPI), Inter-integrated circuit (I2C) interface of STM32F4xx microcontroller

Sources:

1. STMicroelectronics www.st.com : **STM32F101xx, STM32F102xx, STM32F103xx and STM32F105xx/STM32F107xx Reference manual** *(RM0008)*
   STM32F10xxx Cortex®-M3 programming manual *(PM0056)*
   STM32F10xxx Flash memory programming manual *(PM0075)*
   STM32F10xxx XL-density Flash memory programming manual *(PM0068)*

Books Recommended:

1. Ramesh Goanker : **Microcomputer Interfacing,** *McGraw-Hill*

CSE 4142: Microcontrollers Interfacing Lab
Credits: 1 Contact Hours: 26
Year: Fourth Semester: Odd

Prerequisite: CSE2111: Digital System Design, CSE3231: Microprocessor and Microcontroller
Course Type ☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce
Motivation To develop hardware knowledge and microcontroller programming skills for interfacing and apply that knowledge for the real-world applications.

Course Objective:
To enable the interconnection and integration of the physical world with digital devices, STM32F4xx microcontroller, Arduino development board, and to give core concepts of IoT. The hands-on lab work is focused on programming of STM32F4xx microcontroller for interfacing.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **apply** knowledge for real world applications targeting STM32F4xx microcontroller	**Modern tool usage:** (PO5)	Cognitive domain – level 4	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO2	To **apply** knowledge for real world applications using STM32F4xx microcontroller	**Design/development of solutions:** (PO3)	Cognitive domain – level 4	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation

Assessment and Marks Distribution:
Continuous Assessments (CA) (20%)
A comprehensive final exam + Lab note book (70%)
A class participation mark (10%).

Lab Course Contents/List of Experiments:

1. Survey on MCU components GPIO port and registers of STM32F4xx microcontroller.
2. Write program to use GPIO for general purpose I/O, enable GPIO clock (RCC AHB1 Enable Register), Mode Register, Output type and register (Push-pull/Open Drain), Speed Register, Pull-up/Pull-down-Resistor, GPIO port set and Reset register, Alternate function and Alternate Function register
3. Program two GPIO pins for input, Program four GPIO pins for output (use 4 LED)
4. Programming for Nested vectored interrupt controller (NVIC) of STM32F4xx microcontroller,
5. Programming for Analog-to-digital converter of STM32F4xx microcontroller.
6. Programming for Timers
7. Programming for Universal synchronous asynchronous receiver transmitter (USART).
8. Demonstration of Servo motor Control Shield with Auduino

CSE 4151: Computational Geometry
Credits: 3 Contact Hours: 39
Year: Four Semester: Odd

Prerequisite: MATH1221 Co-ordinate Geometry, Vector analysis and Complex Variable
Course Type ☒ Theory ☐ Laboratory work ☐ Project work ☐ Viva Voce
Motivation To know the techniques and concepts needed for the design and analysis of geometric algorithms and data structures.

Course Objective:

The course aims to involve students to rigorous algorithmic analysis for problems in Computational Geometry. And introduce them to applications of Computational Geometry to graphical rendering. Teach them the notions of Voronoi diagrams and Delaunay Triangulations. And develop expected case analyses for linear programming problems in small dimensions.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **understand** the usage of a variety of geometric data structures and algorithms	**Engineering knowledge** (PO1)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☐ Presentation
CO2	To **identify** and **compare** the characteristics and the performance of geometric data structures and algorithms	**Engineering knowledge** (PO1)	Cognitive domain – level 3	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☐ Presentation
CO3	To **apply** fundamental techniques for designing data structures and algorithms suitable for geometric problems	**Engineering knowledge** (PO1)	Cognitive domain – level 4	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☐ Journal paper	☐ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☒ Presentation

Assessment and Marks Distribution:

Students will be assessed on the basis of their overall performance in all the exams, class tests, assignments, and class participation. Final numeric reward will be the compilation of:

Class tests + Assignments due in different times of the semester (20%)
A comprehensive final exam (70%), Total Time: 3 hours.
A class participation mark (10%).

Course Contents:

Introduction: historical perspective, geometric preliminaries. Convex hulls algorithms in 2d and 3d, lower bounds.

Triangulations: polygon triangulations, representations, point-set triangulations, planar graphs;

Voronoi diagrams: construction and applications, variants;

Delayney triangulations: divideand- conquer, flip and incremental algorithms, duality of Voronoi diagrams, min-max angle properties;

Geometric searching: point location, fractional cascading, linear programming with prune and search, finger trees, concatenable queues, segment trees, interval trees;

Visibility: algorithms for weak and strong visibility, visibility with reflections, art-gallery problems;

Arrangements of lines: arrangements of hyperplanes, zone theorems, many-faces complexity and algorithms;

Combinatorial geometry: Ham-sandwich cuts, Helly's theorems, k-sets, polytopes and hierarchies, polytopes and linear programming in d-dimensions, complexity of the union of convex sets, simply connected sets and visible regions;

Sweep techniques: plane sweep for segment intersections, Fortune's sweep for Voronoi diagrams, topological sweep for line arrangements; Randomization in computational geometry: algorithms, techniques for counting; Robust geometric computing; Applications of computational geometry.

Text Book:

1. M. d. Berg, O. Schwarzkopf, M. v. Kreveld and M. Overmars : **Computational Geometry: Algorithms and Applications**, *Springer.*

Books Recommended:

1. F. P. Preparata and M. I. Shamos : **Computational Geometry: An Introduction**, *Springer.*
2. J. O. Rourke : **Computational Geometry in C**, *Cambridge University Press.*

CSE4152: Computational Geometry Lab
Credits: 1 Contact Hours: 26
Year: fourth Semester: Odd

Prerequisite: MATH1221 Co-ordinate Geometry, Vector analysis and Complex Variable
Course Type ☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce
Motivation To develop practical knowledge on the algorithms of Computational geometry.

Course Objective:
This lab course is designed for the students to achieve a hands-on experience on the basic algorithms of Computational Geometry. Theoretical lectures are completed by lab practice where theoretical knowledge is applied.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **implement** the basic algorithms of computational geometry	**Engineering knowledge** (PO1)	Cognitive domain – level 3	☒ Lecture Note ☐ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☐ Assignment ☒ Note book ☐ Presentation
CO2	To **apply** the concept of computational geometry to solve real-world problems.	**Problem analysis** (PO2)	Cognitive domain – level 3	☒ Lecture Note ☐ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☐ Final Exam ☐ Assignment ☒ Note book ☐ Presentation

Assessment and Marks Distribution:

Continuous Assessments (CA) (20%)
A comprehensive final exam + Lab note book (70%)
A class participation mark (10%).

Lab Course Contents/List of Experiments:

1. Convex hull in 2D & 3D: incremental algorithm
2. Convex hull in 3D: divide and conquer algorithm.
3. Intersecting two convex polygons.
4. Linear programming by prune and search.
5. Visibility polygon
6. Binary space partitions and visibility in 2D & 3D
7. Facility location, by prune and search
8. 3D convex hull and Delaunay flips
9. Proximity graphs
10. Computing the Voronoi diagram from the Delaunay triangulation
11. Computing the Voronoi diagram incrementally
12. Counting k-sets

13. Finding ham-sandwich-cuts
14. Arrangements of lines: incremental algorithm
15. Kirkpatrick's algorithm for point location
16. Motion planning for a point robot
17. Shortest path for a polygonal robot
18. Handling an arm robot
19. Intersection of segments (points in general position)
20. Intersection of segments (allowing collinearities)
21. Edge-insertion for optimal triangulation

CSE4161: Digital Image Processing
Credits: 3 Contact Hours: 39
Year: Fourth Semester: Odd

Prerequisite: CSE3221 Digital Signal Processing
Course Type ☒ Theory ☐ Laboratory work ☐ Project work ☐ Viva Voce
Motivation To develop basics knowledge on Digital Image Processing

Course Objective:
This course introduces the basic concepts and methodologies of digital image processing. The covered topics include image representation and storage, image enhancement, high-dimensional spectral analysis, spatial and frequency domain image filtering, binary image processing, edge detection, image segmentation, feature extraction and image compression. It also familiarizes image analysis techniques and concepts.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **describe** the terminology and concept of digital image processing	**Engineering knowledge** (PO1)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☐ Journal paper	☒ Class Test ☐ Final Exam ☐ Assignment ☐ Participation ☐ Presentation
CO2	To **apply** image enhancement and filtering techniques	**Problem analysis** (PO2)	Cognitive domain – level 4	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☐ Journal paper	☐ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☐ Presentation
CO3	To **interpret** image segmentation and restoration	**Engineering knowledge** (PO1)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☐ Journal paper	☐ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☐ Presentation
CO4	To **apply** morphological processing in pattern recovery	**Problem analysis** (PO2)	Cognitive domain – level 4	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☐ Journal paper	☒ Class Test ☐ Final Exam ☒ Assignment ☐ Participation ☐ Presentation

Assessment and Marks Distribution:
Students will be assessed on the basis of their overall performance in all the exams, class tests, assignments, and class participation. Final numeric reward will be the compilation of:

Class tests + Assignments due in different times of the semester (20%)
A comprehensive final exam (70%), Total Time: 3 hours.
A class participation mark (10%).

Course Contents:

Introduction and Fundamental to Digital Image Processing: What is Digital Image Processing, Origin of Digital Image Processing, Examples that use Digital Image Processing, Fundamental steps in Digital Image Processing, Components of Digital Image Processing System, Image sensing and acquisition, Image sampling, quantization and representation, Basic relationship between pixels.

Image Enhancement in the Spatial Domain & Frequency domain: Background, Basic gray level transformation, Histogram processing, Basics of spatial filtering, Smoothing and Sharpening Spatial filters, Introduction to Fourier Transform and the Frequency Domain, Discrete Fourier Transform. Smoothing and Sharpening Frequency-Domain filters.

Image Restoration: Image Degradation/Restoration Process, Noise models, Restoration in presence of noise, Inverse Filtering, Minimum Mean Square Filtering, Geometric mean filter, Geometric transformations.

Image Segmentation: Detection of Discontinuities, Edge linking and boundary detection, Thresholding, Region oriented segmentation.

Morphological image processing: Preliminaries, Dilations and Erosion, opening and closing, Some basic morphological algorithms.

Image Compression: Fundamentals, Image compression models, Error free compression, Lossy compression.

Color Image Processing: Color Fundamentals, Color models, Basis of full color image processing, Color transformations.

Application of image processing in computer vision: Pattern representation, image feature representation pattern classification.

Text Book:

1. Rafael C. Gonzalez and Richard E. Woods : **Digital Image Processing (*4th Edition*),** *Prentice Hall.*

Books Recommended:

1. David Vernon : **Machine Vision: Automated Visual Inspection and Robot Vision,** *Prentice Hall.*
2. William K. Pratt : **Digital Image Processing: PIKS Scientific Inside, 4th Edition,** *John Wiley.*
3. Pitas Ioannis : **Digital Image Processing Algorithms and Applications,** *John Wiley*

CSE4162: Digital Image Processing Lab
Credits: 1 Contact Hours: 26
Year: Fourth Semester: Odd

Prerequisite: CSE3221 Digital Signal Processing
Course Type ☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce
Motivation To develop practical knowledge on Digital Image Processing

Course Objective:
The lab work is not a course in system or processing programming. Its objective is to get you into image processing as soon as possible and acquire an understanding of the purpose and result of selected processes of digital image processing. The work in the course will consist of several homework assignments. In general, the practice with assignment makes this course more understandable to the students. The lab assignments must be done individually and should be submitted on or before the specified deadline.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **demonstrate** basic operations for image conversion	**Modern tool usage** (PO5)	Cognitive domain – level 2	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO2	To **explain** the different point processing operations for image enrichment	**Problem analysis** (PO2)	Cognitive domain – level 2	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO3	To **evaluate** the performance of different filtering techniques for image enhancement	**Design/Development of solutions** (PO3)	Cognitive domain – level 1	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation

Assessment and Marks Distribution:

Continuous Assessments (CA) (20%)
A comprehensive final exam + Lab note book (70%)
A class participation mark (10%).

Lab Course Contents/List of Experiments:

1. Introduction to Matlab to implement digital image processing techniques.
2. Sampling, Resampling, quantization, histogram processing, different arithmetic operations on image.
3. Implementation of basic intensity transformation with point processing techniques and their related functions; concept of spatial filtering, image smoothing and sharpening using different filters.
4. Image enhancement in frequency domain including Fourier transform, Discrete Fourier transform and different high pass filter & low pass filters.
5. Implementation of different edge detection methods and image segmentation.
6. Restoration of image degradation, imposing noise on image, remove noise using filters and measure of filter performance in image enhancement using quantitative approach e.g. PSNR.
7. Implementation of different morphological operations and their application in image enhancement.
8. Comparative study on different algorithms on image compression.

Text Book:

1. Rafael C. Gonzalez and Richard E. Woods : **Digital image processing using Matlab**

Books Recommended:

1. William K. Pratt : **Digital Image Processing: PIKS Scientific Inside, 4th Edition,** *John Wiley.*
2. Pitas Ioannis : **Digital Image Processing Algorithms and Applications,** *John Wiley*

CSE4171: Computer Simulation and Modeling
Credits: 3 Contact Hours: 39
Year: Four Semester: Odd

Prerequisite: MATH2231: Numerical methods

Course Type ☒ Theory ☐ Laboratory work ☐ Project work ☐ Viva Voce

Motivation To develop fundamental knowledge of computer simulation and modelling and understand its necessity and applicability

Course Objective:

In many real-life scenarios, it is not possible to apply a new hypothesis directly to the system. In that situation simulation is the only solution to test the new ideas' usability or applicability. For situations, where complex mathematical computation is needed, computer-generated simulation is the must. Modeling real-world objects in a computing system are an integral part of computer simulation. Bing a CS graduate, each student must have some core knowledge about computer simulation and modeling. And they must be familiar with some popular simulation tools too.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **explain** the different terminologies related with computer modeling and simulation.	**Engineering knowledge** (PO1) **Problem analysis** (PO2)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☐ Participation ☒ Presentation
CO2	To **analyze** the real-life problems and create different types of model of it.	**Problem analysis** (PO2)	Cognitive domain – level 3	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☒ Assignment ☐ Participation ☒ Presentation
CO4	To **produce** the computer simulated solution for a given real world scenario.	**Modern tool usage** (PO5)	Cognitive domain – level 1	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☒ Journal paper	☒ Class Test ☒ Final Exam ☒ Assignment ☒ Participation ☐ Presentation

Assessment and Marks Distribution:

Students will be assessed on the basis of their overall performance in all the exams, class tests, assignments, and class participation. Final numeric reward will be the compilation of:

Class tests + Assignments due in different times of the semester (20%)
A comprehensive final exam (70%), Total Time: 3 hours.
A class participation mark (10%).

Course Contents:

Simulation methods: Introduction to Simulation, Random number generator, analogue simulation of continuous system, Discrete system simulation, Simulation of a pert network, Statistical analysis of result, Validation and verification techniques, Application of simulation to problems e.g. business, operation research, operating system, Computer design and Introduction to simulation packages.

Modeling: Introduction to modeling techniques, Problems, models and systems, Modeling concepts, Logic for (conceptual) modeling, Logic programming for conceptual modeling, Concepts of relational modeling and its practice. Some practical modeling e.g. Relational Database modeling, Different methods for Curves and surface modeling, Fractals, Polyhedral modeling with Euler's formula, Advanced modeling, Procedural models. Case Study: Simulation and Modeling software: Blender, Python and OpenCV or Classic C/C++ GD Library.

Text Books:

1. J. A. Spriet : **Computer Aided Modelling & Simulation**, *Academic Press, Inc. Orlando, FL, USA.*
2. Richard Lehman : **Computer Simulation and Modeling**, *Lawrence Erlbaum Associates Publishers.*
3. G. Cordon : **System Simulation**, Prentice Hall.

CSE4172: Computer Simulation and Modeling Lab
Credits: 1 Contact Hours: 2G
Year: Four Semester: Odd

Prerequisite: MATH2231: Numerical methods

Course Type ☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce

Motivation To put the knowledge gained from CSE4131: Computer Simulation and Modelling theory course into practice and deepen the understandings more.

Course Objective:

The lab is primarily based on the theory course "CSE4131: Computer Simulation and Modeling" to practically exercise and implement the knowledge gathered there. General-purpose programming and modeling tools like C/C++/OpenCV, Python and Blender are utilized throughout.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **implement** random number sequence generators and carryout different type of tests on the generated sequences.	**Modern tool usage** PO5	Cognitive domain – level 4	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO2	To **use** different types of data scrapping techniques to gather information from the open web for analysis.	**Modern tool usage** PO5	Cognitive domain – level 4	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual`	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO3	To **produce** different real-world object models that are ready for use in simulation.	**Modern tool usage** PO5	Cognitive domain – level 4	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☐ Web Material ☒ Lab Manual`	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation

Assessment and Marks Distribution:

Continuous Assessments (CA) (20%)
A comprehensive final exam + Lab note book (70%)
A class participation mark (10%).

Lab Course Contents/List of Experiments:

1. Implementation of different random number generation algorithms
2. Implementation of different categories of tests for checking the properties of randomness of any given number sequence
3. Implementation of simple to complex net crawlers and bot users for different forms of web scrapping scenarios;
4. Conceptualization and materialization of different real-world object models with simple physics.

Text Books:

1. Narsingh Deo : **System Simulation with Digital Computer**, *Englewood Cliffs, N.J.: Prentice-Hall.*
2. G. Cordon : **System Simulation**, Prentice Hall.
3. Ryan Mitchell : **Web Scrapping with Python**, *O'Reilly Media, Inc.*
4. John M. Blain : **The Complete Guide to Blender Graphics Computer Modeling & Animation**, *CRC Press.*

CSE4181: User Interface and User Experience Design (UI/UX Design)
Credits: 3 **Contact Hours:** 39
Year: Four **Semester:** Odd

Prerequisite: None
Course Type ☒ Theory ☐ Laboratory work ☐ Project work ☐ Viva Voce
Motivation To accrue adequate knowledge about designing artifacts that allow the users to meet their needs in the most effective efficient and satisfying manner.

Course Objective:
The course introduces the novice to a cycle of discovery and evaluation and a set of techniques that meet the user's needs. The course mantra is that "Design is a systematic and data driven process." Students will gain an understanding of the critical importance of user interface and user experience design. They will also learn industry-standard methods for how to approach the design of a user interface and user experience along with key theories and frameworks that underlie the design of most interfaces you use today. This course will demonstrate how the costs of bad design can often be severe (in user experience, money, and even human lives).

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **create** new design from scratch	**Design/development of solutions** (PO3)	Cognitive domain – level 5	☒ Lecture Note ☐ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☐ Class Test ☐ Final Exam ☒ Assignment ☒ Participation ☒ Presentation
CO2	To **judge** existing design and suggest improvements	**Design/development of solutions** (PO3)	Cognitive domain – level 6	☒ Lecture Note ☐ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☐ Class Test ☐ Final Exam ☒ Assignment ☒ Participation ☐ Presentation

Assessment and Marks Distribution:
Students will be assessed on the basis of their overall performance in all the exams, class tests, assignments, and class participation. Final numeric reward will be the compilation of:

Class tests + Assignments due in different times of the semester (20%)
A comprehensive final exam (70%), Total Time: 3 hours.
A class participation mark (10%).

Course Contents:
UI Introduction: UI Design and Why it Matters, User Interface Hall of Fame / Shame, Case Study
UI Design Process: Design Process Introduction, Designing to Address a Problem w/o Solution Ideas, Designing for a known solution direction, Designing to iterate on/improve an existing solution, Common Elements, Usability Engineering and Task-Centered Approaches, Use Cases, Personas, Tasks, and Scenarios, Design-Centered Approaches, Design-Centered Methods & When They Work Best, Pulling it all Together: Best from Each; Practical Techniques

Psychology and Human Factors for UI Design: Fitts' Law, Short- and long-term memory, attention, Perception and visualization, hierarchy, Mistakes, Errors, and Slips, Conceptual models, The Gulf of Execution and the Gulf of Evaluation, Design Principles: Visibility, Feedback, Mappings, Constraints, Interacting beyond individuals

(social psychology), High-Level Models: Distributed Cognition, Activity Theory, Situated Action, Interface Critiques, Overview and Properties of a Good Critique, Psychology and Human Factors: Shortcuts to Understanding Your Users

User Experience Design Overview: Important Terms and Concepts, Features of Good Design, User Engagement Ethics, General Resources, Overview of User Experience Design

Requirement Gathering: Types of Users and Types of Data, Discovery Technique, Naturalistic Observation, Survey, Focus Group, Interview, User Results, Presenting Task Findings, Elements of Requirement Gathering Designing Alternatives: Review of Design Goals, Design Alternatives. Prototyping: Resources and Tools for Prototyping, Resources for Prototyping. Evaluation: Resources for Evaluation

Text Book:

1. Donald A. Norman : **The Design of Everyday Things**

Books Recommended:

1. Jesse James Garret : **The Elements of User Experience**
2. Jenifer Tidwell : **Designing Interfaces: Patterns for Effective Interaction Design**

CSE4182: User Interface and User Experience Design (UI/UX Design) Lab

Credits: 1 **Contact Hours:** 26
Year: Four **Semester:** Odd

Prerequisite None

Course Type ☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce

Motivation To accrue adequate practical knowledge about designing artifacts that allow the users to meet their needs in the most effective efficient and satisfying manner.

Course Objective:

The course introduces the novice to a cycle of discovery and evaluation and a set of techniques that meet the user's needs. The course mantra is that "Design is a systematic and data driven process." Students will gain an understanding of the critical importance of user interface and user experience design and practically demonstrate them. They will also work on industry-standard methods for how to approach the design of a user interface and user experience along with key theories and frameworks that underlie the design of most interfaces you use today.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	Create new UI design from scratch	**Design/development of solutions** (PO3)	Cognitive domain – level 5	☒ Lecture Note ☐ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO2	Gain practical experience in UX Design	**Design/development of solutions** (PO3)	Cognitive domain – level 6	☒ Lecture Note ☐ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation

Assessment and Marks Distribution:

Continuous Assessments (CA) (20%)
A comprehensive final exam + Lab note book (70%)
A class participation mark (10%).

Lab Course Contents/List of Experiments:

1. Case Studies
2. UI Disasters, including GPS fails
3. Corporate Value: Citibank ATM
4. Microsoft Office 2007 Ribbon
5. International Children's Digital Library
6. Taxes and Tickets
7. AirBnB vs. CouchSurfing
8. Task:

1) User research and ideation
First prototype
Cognitive Walkthrough and heuristic evaluation
Second prototype and user test plan
User test
Carry out a significant, realistic project from start to finish, applying UX methods in an iterative design process
   a. Produce a portfolio project that can show employers what you can do

Text Book:

1. Donald A. Norman : **The Design of Everyday Things**

Books Recommended:

1. Jesse James Garret : **The Elements of User Experience**
2. Jenifer Tidwell : **Designing Interfaces: Patterns for Effective Interaction Design**

CSE4191: Blockchain
Credits: 3 Contact Hours: 39
Year: Fourth Semester: Odd

Prerequisite: None

Course Type ☒ Theory ☐ Laboratory work ☐ Project work ☐ Viva Voce

Motivation It is ideal for programmers and designers involved in developing and implementing blockchain applications, and anyone who is interested in understanding its potential.

Course Objective:

This Course introduces blockchain, a revolutionary technology that enables peer-to-peer transfer of digital assets without any intermediaries, and is predicted to be just as impactful as the Internet. More specifically, it prepares learners to program on the Ethereum blockchain.

The course covers a range of essential topics, from the cryptographic underpinnings of blockchain technology to enabling decentralized applications on a private Ethereum blockchain platform.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **recognize** foundational concepts of blockchain, .	**Engineering knowledge** (PO1)	Cognitive domain – level 3	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☒ Participation ☐ Presentation

CO2	To **design**, code, deploy and execute a smart contract.	**Design/development of solutions** (PO3)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☐ Assignment ☒ Participation ☐ Presentation
CO3	To **design** and develop end-to-end decentralized applications (Dapps).	**Design/development of solutions** (PO3)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ Class Test ☒ Final Exam ☒ Assignment ☒ Participation ☒ Presentation

Assessment and Marks Distribution:
Students will be assessed on the basis of their overall performance in all the exams, class tests, assignments, and class participation. Final numeric reward will be the compilation of:

Class tests + Assignments due in different times of the semester (20%)
A comprehensive final exam (70%), Total Time: 3 hours.
A class participation mark (10%).

Course Contents:
Blockchain Defined: Practitioner's Perspective, ConsenSys, Blockchain Defined: Bitcoin & Blockchain, Blockchain Structure, Basic Operations, Beyond Bitcoin. Ethereum Blockchain: Smart Contracts, Practitioner's Perspective: The Enterprise, Ethereum Structure, Ethereum Operations, Incentive Model.

Algorithms & Techniques: Public-Key Cryptography, Hashing, Transaction Integrity, Securing Blockchain Trust Essentials: Decentralized Systems, Consensus Protocol, Practitioner's Perspective: Decentralized Governance, Robustness, Forks. Smart Contract Basics: Smart Contract Basics: Why Smart Contracts? Smart Contracts Defined, Processing Smart Contracts, Deploying Smart Contracts.

Solidity: Solidity: Structure, Basic Data Types & Statements (Bidder Data & Functions Demos), Specific Data Types, Data Structures, Access Modifiers & Applications. Putting it all Together: Developing Smart Contracts, Time Elements, Validation & Test, Client Applications, Practitioner's Perspective: Shared Rules & Trust Best Practices: Evaluating Smart Contracts, Designing Smart Contracts Remix Web IDE

Decentralized Applications (Dapps): Blockchain Server, Dapp Defined, Ethereum APIs, Practitioner's Perspective: Public Network Architecture. Truffle Development: Truffle IDE, Test-Driven Development, Web Interface & Testing.

Design Improvements: Solidity Features, Event Handling, Oraclize. Application Models & Standards: Dapp Models, Dapp Standards. Permissioned Blockchains: Hyperledger, Fabric Services, Fabric Model & Functions, Composer, Microsoft Azure.

Decentralized Applications Platforms: Augur, Grid+, Challenges & Solutions: Consensus, Scalability, Privacy & Confidentiality, Escrow & Multi-sig. Alternative Decentralized Solutions: Interplanetary File System, Hashgraph, Social Imperative, Practitioner's Perspective: Market Adoption

Text Book:

1. Tiana Laurence : **Blockchain for Dummies**, Wiley, Year: 2019

Books Recommended:

1. Horst Treiblmaier, Roman Beck : **Business Transformation through Blockchain**: *Volume I & II, Palgrave Macmillan, Year: 2019*
2. Jai Singh Arun; Genarro Cuomo; Nitin Gaur : **Blockchain for Business**, *Addison-Wesley Professional, Year: 2019*
3. Xiwei Xu, Ingo Weber, Mark Staples : **Architecture for Blockchain Applications**, *Springer, Year: 2019*

CSE4192: Blockchain Lab
Credits: 1 Contact Hours: 26
Year: Fourth Semester: Odd

Prerequisite: None
Course Type ☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce
Motivation It is ideal for programmers and designers involved in developing and implementing blockchain applications, and anyone who is interested in understanding its potential.

Course Objective:
This Course introduces blockchain, a revolutionary technology that enables peer-to-peer transfer of digital assets without any intermediaries, and is predicted to be just as impactful as the Internet. More specifically, it prepares learners to program on the Ethereum blockchain.
The course covers a range of essential topics, from the cryptographic underpinnings of blockchain technology to enabling decentralized applications on a private Ethereum blockchain platform.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
CO1	To **create** nodes on a private Ethereum blockchain	**Design/development of solutions** (PO3)	Cognitive domain – level 6	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO2	To **design** develop and test a smart contract	**Design/development of solutions** (PO3)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation
CO3	To **design** and develop a decentralized application	**Design/development of solutions** (PO3)	Cognitive domain – level 5	☒ Lecture Note ☒ Text Book ☒ Audio/Video ☒ Web Material ☐ Journal paper	☒ CA ☒ Final Exam ☒ Assignment ☒ Note book ☒ Presentation

Assessment and Marks Distribution:
Continuous Assessments (CA) (20%)
A comprehensive final exam + Lab note book (70%)
A class participation mark (10%).

Course Contents:
1. create nodes on a private Ethereum blockchain, create accounts, unlock accounts, mine, transact, transfer Ethers, and check balances.
2. To design, develop and test a smart contract for a problem using Solidity language and Remix IDE.
3. To apply incremental development method and best practices.
4. Design and develop a decentralized application (Dapp) using Truffle Integrated Development Environment (IDE)
5. Explain and apply the concepts of test-driven development
6. Steps in Dapp development using Truffle IDE.
7. Install IPFS on your compute machine
8. Use command line interface (CLI) to IPFS to add a file on IPFS

9. Use the web interface to the IPFS to view your peers and add (drag and drop) files to IPFS
10. Add a folder of files to IPFS using the web interface to the IPFS, view the hash of the folder and the files on the CLI
11. Use IPFS to share files with others
12. The base-58 addressing scheme of IPFS objects

Text Book:

1. Tiana Laurence : **Blockchain for Dummies**, *Wiley, Year: 2019*

Books Recommended:

1. Horst Treiblmaier, Roman Beck : **Business Transformation through Blockchain**: Volume I & II, *Palgrave Macmillan*
2. Jai Singh Arun; Genarro Cuomo; Nitin Gaur : **Blockchain for Business**, *Addison-Wesley Professional*
3. Xiwei Xu, Ingo Weber, Mark Staples : **Architecture for Blockchain Applications**, *Springer*

CSE4192: Thesis/Project (Part-I)
Credits: 1 Contact Hours: 26
Year: Fourth Semester: Even

Prerequisite: None
Course Type ☐ Theory ☐ Laboratory work ☒ Project work ☒ Viva Voce
Motivation To design a develop a project from their knowledge they have acquired from their undergraduate program

Course Objective:
Each student has to complete one project in the combined duration of two semesters of Part-IV. In odd semester course CSE 4192 (Part-I), a student has to make a proposal defense at the end of the semesters. The defensed project has to be completed in the continuation course CSE 4292 (Part-II) in even semester of Part-IV. This course is designed for the students to achieve their skills function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

CO No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Assessment tools
CO1	To **Construct** a real world project based on their acquired knowledge	**Individual work and teamwork** (P09)	Cognitive domain – level 1	☐ CA ☒ Project presentation ☐ Assignment ☒ Project paper ☒ Oral Presentation

Assessment:
25 Marks (35% Internal Examiner, 35% External Examiner, 30% Presentation and Oral).